

## PART I (30 MARKS)

(Attempt *all* questions from this *Part*)

### Question 1

Study the extract of the **Survey of India** Map Sheet No. **G43S10** and answer the following questions:

- (i) (a) Give the *six-figure* grid reference for spot height **.437** in the *southeastern* part of the map extract. [2]
- (b) Aman, a local craftsman, wants to sell his product to a large gathering to earn money instantly.
1. Which settlement should he go to?
2. Name the month in which he should visit the settlement named by you above.
- (ii) (a) On which bank of the ***Sukli nadi*** is the settlement of ***Pamera*** situated? [2]
- (b) Identify the **settlement pattern** seen in the grid square **6732**.
- (iii) Roshan, while travelling from **Sirori** to **Tokra**, observed that the surface water sources are dry except for the reservoir near **Tokra**. [2]
- (a) What according to you may be the reason for the presence of water in the reservoir in **6832** near Tokra settlement?
- (b) State whether this reservoir is *artificial* **or** *natural* source of water.
- (iv) (a) What is the *direct distance* between **Gulabganj** to **Pithapura**? Give your answer in **kilometre**. [2]
- (b) Mention the *compass direction* of **Pithapura** **from** **Gulabganj**.
- (v) (a) Name **two** *man-made features* seen in the grid square **6534**. [2]
- (b) Define *contour interval*.

## MARKING SCHEME

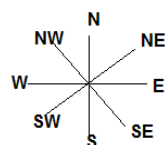
|       |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>(a) <math>67\frac{2}{3}</math> <math>29\frac{4}{5}</math></p> <p>(b) 1. Malagaon / Pamera (<i>any one</i>)</p> <p>2. March</p>                                                                                                                                                                                                                                                                                |
| (ii)  | <p>(a) Right bank/ Right side/ Right</p> <p>(b) Scattered / dispersed</p>                                                                                                                                                                                                                                                                                                                                        |
| (iii) | <p>(a) There is a presence of embankment / sluice. / dam/ storing water for irrigation./ only storing water/ for growing crops.</p> <p>(b) It is an artificial reservoir./ Man-made</p>                                                                                                                                                                                                                          |
| (iv)  | <p>(a) 2.3 km- range between 1.5 km-3.5 km</p> <p>(b) North east / NE/ NNE</p>                                                                                                                                                                                                                                                                                                                                   |
| (v)   | <p>(a) Post office, permanent hut, metalled road, temple, dispensary, lined perennial well./well/Settlement/hut/ road/ National Highway/ Cart track/ Chattri/ Power line/ Unsurveyed poles houses.</p> <p>(b) It is the difference in value between two successive/consecutive contour lines.<br/>Vertical distance/gap between two successive contours/Difference in height between two successive contour.</p> |

## Comments of Examiners

- (i) (a) Most of the candidates were unable to figure out the third and the sixth figure. Some candidates had written the sixth number of the northing with the 'eastings two-digit number' and vice versa. The three digits of the northings were written first, followed by the eastings' three digits.
- (b) Several candidates attempted this part incorrectly. In this, some candidates did not know or understand what a large gathering is. They mentioned the different names of the villages instead of the month as asked in the question. Majority of the candidates did not know they had to write the name of the settlement; instead, they mentioned the type of settlement, i.e. nucleated. However, a very few candidates mentioned the month correctly despite the first part being attempted incorrectly.
- (ii) (a) A large proportion of the candidates mentioned the concept of riverbank inaccurately, as they were unable to figure out the bank of Sukhli Nadi with reference to the Pamera settlement. In addition, several candidates mentioned 'direction' or 'northern'/'southern side' instead of the 'right bank'. Thus, there are two conceptual gaps identified at two levels: (a) recognising features and their placement on the toposheet, (b) determining the stream's direction.
- (b) Most of the candidates were able to figure out the settlement pattern, but for a few of the candidates, the identification of the settlement pattern was incorrect. They were confused about the settlement pattern and drainage pattern.
- (iii) (a) Most of the candidates gave the wrong answer as they wrote due to streams from contours.
- (b) Majority of the candidates mentioned an incorrect answer as 'natural'.
- (iv) (a) Most of the candidates gave the wrong answer. In addition, some candidates did not mention the units 'km'. The other few candidates mentioned '5.9 kms'. However, the correct answer ranged between 1.5 km to 3.5 km.
- (b) A large number of the candidates did not attempt this part correctly, as they were unclear about the compass directions.

## Suggestions for teachers

- Clearly explain the concept of a 6-figure grid reference. Draw attention to Eastings and Northings, and provide practice exercises where Eastings are mentioned first, followed by Northings.
- Explain how to interpret a toposheet effectively.
- Emphasise the importance of people gathering at a particular place. Provide hints and guide learners to understand the reasons behind gatherings at specific places and times of the year.
- Encourage reading the question multiple times to understand what is being asked.
- Teach the basic concept of the flow of a river, emphasising the need to face the direction of the river's flow while interpreting features.
- Clarify the concept of topography at the foundational level.
- Help visualise an artificial or natural reservoir, highlighting the point of embankment through detailed explanation and description.
- Explain that any embankment or structure which obstructs or stores water is considered an artificial construction.
- Demonstrate how to measure distances using a scale—both direct distances (as the crow flies) and along the actual path.
- Explain compass directions and their subdivisions, introducing cardinal directions first, followed by intermediate directions.



- Bring clarity about man-made (cultural) and natural features. Correlate their identification with the conventional symbols provided at the end of the map extract.

- (v) (a) Majority of the candidates mentioned the man-made features correctly; however, a few of the candidates, to whom the topography was not clear, gave the man-made features which was not even given in the specified area. Many responses were given, such as hospitals, houses, and a police station, even though they were not present.
- (b) A large number of candidates confused 'definition of Contour Interval' with 'Vertical interval'. In addition, many responses mentioned 'the contour interval is 20 m. These responses indicated that candidates did not have some fundamental concepts that are crucial for comprehending slope and landforms.

### Suggestions for teachers

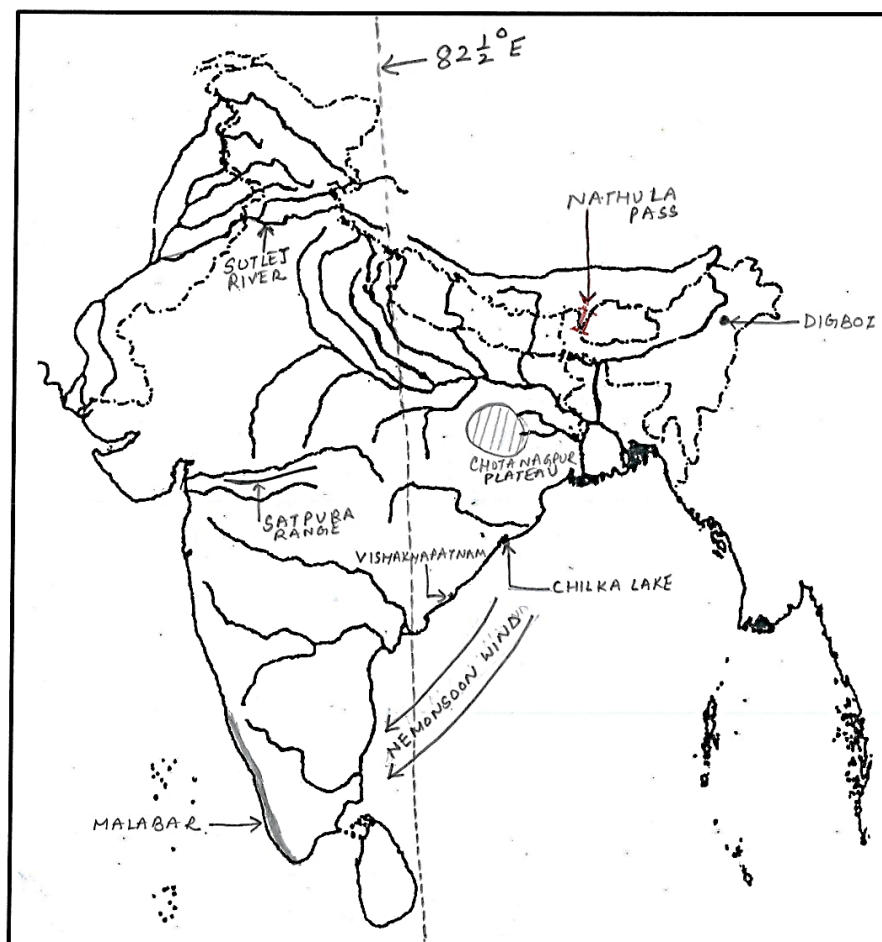
- Explain the features of natural and man-made elements. Show live examples on the toposheet, such as natural features, a man-made tank with an embankment, and a prominent surveyed tree.
- Discuss the geographical definitions of contour, contour interval, and vertical interval in class.
- Guide learners to develop a comprehensive understanding of the terrain, landforms, landscapes, and human settlements in an area by studying conventional symbols, scale, contour lines, distance, and direction.
- Develop analytical reasoning to understand how different attributes, such as the direction of a stream and slope, are interconnected.
- Promote analytical skills through hands-on activities that involve analysis and application-based questions related to toposheet features.
- Ensure correct understanding of the definitions of RF, contours, contour interval, black broken lines, black curved lines, black vertical lines, and black horizontal lines.

## Question 2

On the outline map of India provided:

- |        |                                                                                                   |     |
|--------|---------------------------------------------------------------------------------------------------|-----|
| (i)    | Mark and label the <i>Nathu-La pass</i> .                                                         | [1] |
| (ii)   | Shade and name the <i>coastal plain</i> that occupies a large part of <i>Kerala</i> .             | [1] |
| (iii)  | Mark and label <i>Digboi</i> .                                                                    | [1] |
| (iv)   | Mark and label <i>Chilka lake</i> .                                                               | [1] |
| (v)    | Mark with arrows and <i>name the wind</i> that brings rainfall to Tamil Nadu coast during winter. | [1] |
| (vi)   | Mark and label the river <i>Sutlej</i> .                                                          | [1] |
| (vii)  | Mark and label the <i>Standard Meridian of India</i> .                                            | [1] |
| (viii) | Mark and label <i>Satpura range</i> .                                                             | [1] |
| (ix)   | Shade and label the <i>Chota Nagpur plateau</i> .                                                 | [1] |
| (x)    | Mark and label <i>Visakhapatnam</i>                                                               | [1] |

## MARKING SCHEME



### Comments of Examiners

- (i) Most of the candidates did not mark 'Nathu la Pass' correctly. Instead, the candidates marked the 'north of Sikkim' or 'between Nepal and Sikkim'.
- (ii) Several candidates marked it on the 'Konkan Coast' instead of 'Kerala below river Kaveri'. However, some candidates marked correctly but mentioned the incorrect name. The remaining few candidates marked it on the 'Eastern side'.
- (iii) A large number of candidates marked 'Digboi' incorrectly.
- (iv) Most of the candidates marked 'Chilka Lake' correctly.
- (v) Almost half of the candidates answered this question incorrectly, while some of those who selected the correct option labelled it inaccurately.

### Suggestions for teachers

- Show live examples on the digital screen and mark their exact locations on the map.
- Provide tips such as coasts are on the eastern and western sides.
- Show the exact location of Digboi on the map in class.
- Indicate on the map, under 'Mahanadi', which coastline extends inland.
- Display on the board that during winter, Tamil Nadu receives rainfall.
- Help students mark the Indian and Northern rivers on the map.
- Use simple words like black vertical line for longitude and black horizontal line for latitude to explain the differences.

- (vi) A large proportion of candidates marked 'Sutlej' incorrectly. Several candidates marked 'Beas and Sutlej', and the remaining few of the candidates marked the Indus.
- (vii) Most of the candidates did not mark it correctly, as 'Standard Meridian' was marked as 'Tropic of Cancer'. In some of the responses, incorrect labelling was identified.
- (viii) Majority of the candidates marked 'Satpura range' correctly; however, some candidates marked it on 'the northern side of river Narmada'.
- (ix) Some candidates marked, 'Chota Nagpur Plateau' correctly. The remaining candidates were unclear about the exact location.
- (x) Visakhapatnam was not marked properly by most of the candidates. The candidates marked it close to 'Godavari' or close to 'north of the exact location'.

### Suggestions for teachers

- Provide a hint stating that the Satpuras are located between the Narmada and Tapi.
- Indicate that a feature may be marked before the Damodar River, for example, with a circle.
- Demonstrate the projected coastline into the sea above the Godavari River in class.
- Give practice in identifying the location of various features, such as rivers relative to the states they flow through, river direction in relation to topography, cities along rivers, and the coastal plain in relation to coastal areas.
- Build a meaningful understanding of location by connecting different geographical features on the map—for instance, determining whether a riverine port like Kolkata is on the coastline or inland, or identifying the location of river deltas along the eastern coast by examining the outline of the eastern coastal plain.

## Question 3

[10]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only.)

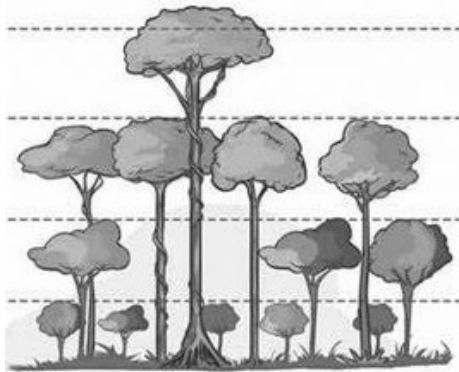
- (i) **Assertion (A):** Mumbai receives more rainfall than Pune.

**Reason (R):** Mumbai gets rain from Arabian Sea branch of southwest monsoon.

- (a) Both (A) and (R) are true and (R) is the correct explanation for (A).
  - (b) Both (A) and (R) are true and (R) is not the correct explanation for (A).
  - (c) (A) is true and (R) is false.
  - (d) (A) is false and (R) is true.
- (ii) Which of the following *pairs of soils* displays **similar characteristics** in terms of their **colour and texture**?
- (a) Alluvial soil and laterite soil
  - (b) Black soil and red soil

- (c) Red soil and laterite soil
- (d) Alluvial soil and black soil

(iii) The *type of natural vegetation* shown in the picture given below is found in:



- (a) Tropical desert
  - (b) Tropical deciduous
  - (c) Tropical evergreen
  - (d) Littoral forest
- (iv) Which of the following *prevents wastage of water*?
- (a) Canal
  - (b) Well
  - (c) Sprinkler
  - (d) Tube well
- (v) Balaghat : Manganese : : Mumbai High : \_\_\_\_\_
- (a) Copper
  - (b) Coal
  - (c) Iron Ore
  - (d) Petroleum
- (vi) A *microbiological process* that loosens the bark of a crop and facilitates *removal of fibre* from the stalk:
- (a) Rolling
  - (b) Ratooning
  - (c) Retting

- (d) Fermentation
- (vii) Which industry amongst the following has revolutionised the lifestyle of the Indian masses in recent years?
  - (a) Cotton Textile
  - (b) Electronic
  - (c) Iron and Steel
  - (d) Sugar
- (viii) Which of the following household products is associated with the petrochemical industry?
  - (a) Glass bottle
  - (b) Wooden chopping board
  - (c) Kitchen knife
  - (d) Plastic mug
- (ix) Raman, a dairy farmer in Ludhiana, supplies milk within the city on a daily basis. Which mode of transport is he most likely to use?
  - (a) Railway
  - (b) Roadway
  - (c) Airway
  - (d) Waterway
- (x) The collection of discarded materials and converting them into useful products is:
  - (a) Recycling
  - (b) Processing
  - (c) Reducing
  - (d) Composting

## MARKING SCHEME

|        |                                                                               |
|--------|-------------------------------------------------------------------------------|
| (i)    | (b) Both (A) and (R) are true but (R) is not the correct explanation for (A). |
| (ii)   | (c) Red soil and laterite soil                                                |
| (iii)  | (b) Tropical deciduous/ (c) Tropical evergreen                                |
| (iv)   | (c) Sprinkler                                                                 |
| (v)    | (d) Petroleum                                                                 |
| (vi)   | (c) Retting                                                                   |
| (vii)  | (b) Electronic                                                                |
| (viii) | (d) Plastic mug                                                               |
| (ix)   | (b) Roadway                                                                   |
| (x)    | (a) Recycling                                                                 |

### Comments of Examiners

- (i) Majority of the candidates did not understand the question and mentioned option (a) as the answer. The candidates were not able to apply the knowledge as they struggled to grasp the significance of the Western Ghats as an obstacle to the Arabian Sea branch of the Monsoon.
- (ii) Almost all the candidates attempted this part correctly.
- (iii) A large number of candidates did not recognise the layered structure of Tropical Evergreen forests.
- (iv) Several candidates attempted this part correctly.
- (v) Almost all the candidates answered it accurately.
- (vi) A large number of candidates mentioned 'ratooning' as the response.
- (vii) Majority of the candidates mentioned option (c) as the answer that reflected a gap in understanding of the meaning.
- (viii) Almost all the candidates attempted this part correctly.
- (ix) No major mistakes were identified in this part of the question.
- (x) A large number of the candidates attempted this part correctly.

### Suggestions for teachers

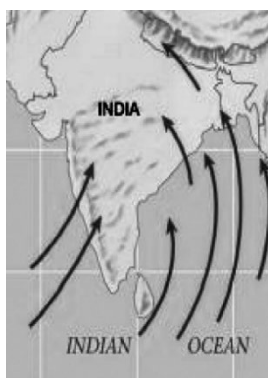
- Guide students to read the question and options carefully to answer correctly.
- Encourage learning of the characteristic features of all soil types.
- Discuss in detail how mountains influence rainfall patterns, using locations like Mumbai and Pune as examples.
- Provide a clear distinction between tropical evergreen and tropical deciduous forests through practice questions.
- Differentiate between types of vegetation in class using proper characteristics and diagrams.
- Explain theory to prevent water wastage effectively.
- Encourage memorisation of the names of places where minerals are mined.
- Familiarise learners with the terms 'Ratooning' and 'Retting' along with their associated crops.
- Discuss the uses of industries and their impacts on people's lives.
- Encourage associating household products with the petrochemical industry.
- Build clarity on the uses of transport and different modes of transport.
- Clarify concepts related to methods of waste disposal.

## PART II – (50 MARKS)

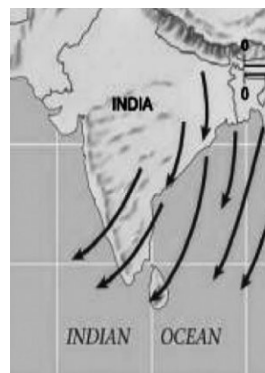
(Attempt *any five* questions from this *Part*.)

### Question 4

- (i) 'Severe cyclonic storms are often formed over the Bay of Bengal during the retreating monsoon season.' [2]
- (a) Name *two* states that are affected by tropical cyclones on the *eastern coast* of India.
- (b) The temperate cyclones cause less damage than the tropical cyclones. Give *any two points* to support this statement.
- (ii) (a) Name the *local wind* that causes rainfall in the states of **Assam** and **West Bengal** in the month of *April*. [2]
- (b) How is this rainfall *beneficial* for the states mentioned above?
- (iii) With reference to **Figure A** and **Figure B** given below, answer the questions that [3]  
follow:



**Figure A**



**Figure B**

- (a) Name the winds shown by arrows in Figure A and Figure B.
- (b) How is the wind shown in Figure A different from the one shown in Figure B?
- (iv) Study the climate data of station X and answer the following questions: [3]

| Month         | Jan  | Feb  | Mar  | Apr  | May  | June | July | Aug  | Sep  | Oct  | Nov  | Dec  |
|---------------|------|------|------|------|------|------|------|------|------|------|------|------|
| Temp (°C)     | 24.0 | 25.4 | 26.7 | 28.3 | 30.0 | 28.9 | 27.6 | 27.4 | 27.3 | 27.1 | 27.0 | 25.0 |
| Rainfall (cm) | 0.25 | 0.25 | -    | -    | 1.75 | 50.2 | 61.0 | 37.0 | 27.0 | 4.75 | 1.50 | -    |

- (a) What is the **annual range of temperature** for station X?
- (b) Which is the **wettest month**?
- (c) Name the **wind** which brings rainfall for station X during the **rainy season**.

## MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>(a) West Bengal / Odisha / Andhra Pradesh / Tamil Nadu <span style="float: right;">(any two)</span></p> <p>(b)</p> <ul style="list-style-type: none"> <li>• Temperate cyclone winds move with less speed/less stormy/no storms.</li> <li>• Temperate Cyclone large area/ beneficial for Rabi Crop.</li> <li>• Temperate cyclone brings in light rainfall /drizzle.</li> <li>• Gentle pressure Gradient.</li> </ul> <p><i>Whereas tropical cyclonic winds move fast /and bring heavy rainfall which leads to a lot of destruction./ Steep pressure gradient affects small area.</i></p> <p style="text-align: right;">(any two)</p>                                                                                                                                                        |
| (ii)  | <p>(a) Kalbaisakhi / Norwester / Bardoli chera <span style="float: right;">(any one)</span></p> <p>(b) It is helpful for growing rice and tea/ helpful for rice production/helpful for tea production / helpful for crop production/ helpful for jute/ paddy, it lowers the temperature and so brings relief.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (iii) | <p>(a) Figure A shows South West monsoon wind / onshore winds/ summer monsoon/approaching monsoon./ advancing monsoon/ sea to land./ Arabian Sea branch/ Bay of Bengal branch.</p> <p>Figure B shows North East monsoon wind /off shore winds/ winter monsoon/retreating monsoon./ Land to sea.</p> <p>(b) Wind seen in Figure A is onshore and the one in Figure B is offshore wind.</p> <p>A-Summer Monsoon, B- Retreating Monsoon</p> <p>Wind seen in Figure A bring rain to most of India and the one in Figure B bring rain to S E coastal areas/ Coromandal Coast/ TN Coast</p> <p>Wind seen in Figure A blows during summer and the one seen in Figure B blows during winter./ Retreating</p> <p>Mention of Month/ Any month between- June to September, and October to February.</p> |
| (iv)  | <p>(a) <math>30 - 24 = 6^{\circ}\text{C}</math></p> <p>(b) July is the wettest month.</p> <p>(c) South west monsoon wind./ Summer monsoon/advancing monsoon. / Approaching Monsoon.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Comments of Examiners

- (i) (a) Majority of the candidates who attempted this question were able to write the two states. However, their responses indicated that they require a more comprehensive understanding of localised weather phenomena.
- (b) A large number of candidates were not able to apply their knowledge of temperate & tropical cyclones from their previous classes.
- (ii) (a) Most of the candidates attempted this part correctly as they were clear about the local winds.
- (b) Several candidates reflected in their responses a well-developed understanding of the benefits of the wind.
- (iii) (a) Most of the candidates recognised the winds.
- (b) A large number of candidates were not able to apply their knowledge.
- (iv) (a) Majority of the candidates answered it accurately. In some instances, the candidates were not able to mention the 'units'; however, in certain cases, calculation errors were found.
- (b) Almost all the candidates answered it correctly.
- (c) Most of the candidates answered this sub-part of the question correctly. Only a few candidates mentioned it as 'Retreating Monsoon'.

## Suggestions for teachers

- Explain cyclones with proper examples, and emphasise to students the importance of reading the question carefully.
- Build students' conceptual clarity between the two types of cyclones: tropical cyclones and temperate cyclones.
- Explain and guide students to learn the names of different local winds and the regions where they occur.
- Clarify the relationship between rainfall and the crops associated with specific rainfall types in different areas.
- Explain and discuss the basic concepts of low pressure and high pressure with students.
- Build clarity on the direction of winds during different monsoon seasons.
- Emphasise the importance of mentioning units in calculation-based questions.
- Guide students to carefully read and interpret data before answering questions.
- Discuss the differences in the duration and period of various types of monsoons.
- Explore local winds in depth, along with broader atmospheric circulation patterns, using case studies on local conditions, formation of winds, and their impact on local economies and livelihoods.
- Draw students' attention to reading and interpreting climatic data, understanding how it is presented through relevant indicators, and extracting information about a location's climate.
- Use visual aids, diagrams, photographs, and stories to enhance understanding of physical and climatic phenomena. For example, employ pictures, diagrams, and videos to explain the formation differences between tropical and temperate cyclones.

## Question 5

- (i) The Ganga Plain is known for its fertile soil. It is often referred to as the food bowl of India. This region supports a variety of crops such as rice, wheat, pulses, jute and sugarcane. [2]
- (a) Name the soil that is found in large parts of Ganga plain.
- (b) Give *two* properties of this soil which helps it to produce a variety of crops.
- (ii) (a) What is meant by *soil erosion*? [2]
- (b) Name *any two* agents of *soil erosion*.
- (iii) Give a geographical reason to explain each of the following: [3]
- (a) **Soil conservation** is the need of the hour.
- (b) **Red soil** has a poor water retention capacity.
- (c) **Laterite soils** are infertile.
- (iv) With reference to **black soil** of India, answer the following questions. [3]
- (a) How is this soil formed?
- (b) How is this soil able to retain moisture for a long period?
- (c) Mention *any two* minerals present in this soil.

### MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>(a) Alluvial soil / Khadar/bangar/new alluvium/old alluvium./ in land alluvium.</p> <p>(b) Alluvial soil is rich in Potash, lime, humus, iron ,magnesium, calcium, aluminium (can be taken as separate points) / it is renewed or replenished or fresh deposition / friable or easily cultivable/moisture retentive/fine textured/responds to fertilizers/loamy./ It is fertile/ deep.</p> <p style="text-align: right;">(<i>any two</i>)</p> |
| (ii)  | <p>(a) It is the removal and / or destruction of top soil due to running water, wind or activities of man./ it is removal of fertile soil/removal of top layer of soil/depletion of top soil/ degradation/ detachment and removal/ detachment and transportation.</p> <p>(b) Running water, wind./ man/ sea waves/ water.</p> <p style="text-align: right;">(<i>any two</i>)</p>                                                                 |
| (iii) | <p>(a) It is needed to preserve and protect the fertile layer so that the productivity of soil is retained./ to preserve soil for feeding growing population/if soil is not conserved it may lead to scarcity of food./ Soil takes a long time to form so it should be consumed.</p> <p>(b) Red soil is porous./coarse texture./ loose soil.</p>                                                                                                 |

|      |                                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | (c) It is due to leaching/ poor in nitrogen, lime, phosphorus, magnesium, organic matter/poor water retention/acidic/poor in silica/poor in nutrients.                                                                                                                            |
| (iv) | (a) It is formed by weathering of lava rocks./disintegration of volcanic/basaltic/extrusive igneous rocks/ Basic igneous solidification of basic lava.<br>(b) It has clayey nature./fine textured/sticky when wet<br>(c) Lime, iron, magnesium, calcium, alumina, potash (LIMCAP) |

(any two)

## Comments of Examiners

- (i) (a) Most of the candidates were able to identify the type of soil found in a large part of the Ganga plain.
- (b) A large number of candidates answered this part correctly. However, for two properties of Alluvial soil, a few candidates mentioned that it contains clay to hold moisture. The candidates found it difficult to grasp the characteristics of various soil types, such as texture, porosity, and colour.
- (ii) (a) It was answered correctly by several candidates.
- (b) Some candidates mentioned Anthropogenic factors as agents of soil erosion, and a few candidates mentioned causes in place.
- (iii) (a) It was answered correctly by most of the candidates.
- (b) Several candidates mentioned that Red Soil contains a high amount of iron oxide; hence, it has low moisture retention capacity.
- (c) Most of the candidates mentioned this answer correctly.
- (iv) (a) In some cases, candidates mentioned that Black Soil is formed by the denudation of lava rocks.
- (b) Majority of the candidates were able to answer it correctly.
- (c) A large proportion of the candidates were able to mention two minerals present in Black Soil.

## Suggestions for teachers

- Explain the formation and distribution of each type of soil using maps.
- Discuss the properties and composition of different soils as per the syllabus.
- Build clarity on the characteristics of specific soils, such as Laterite soil.
- Explain soil erosion, its causes by different agents, and the importance of soil conservation methods.
- Use practical examples: encourage students to collect soil samples and compare them based on formation, properties, and composition.
- Promote classroom discussions where students share observations of physical and cultural phenomena related to soil.

## Question 6

- (i) *India has lost 2.33 million hectares of tree cover since the year 2000. This is equivalent to a six percent decrease in tree cover during this period.* [2]

*Source: The Indian Express, Date: 12.04.2024*

- (a) Mention *any two* causes of the decrease in the forest cover of India.
- (b) Name a *conservation method* that may be used to help in increasing the forest cover.

- (ii) Mention the *climatic condition* suitable for the development of tropical evergreen forest. [2]
- (iii) State *any three* differences between tropical deciduous and tropical desert vegetation. [3]
- (iv) (a) Give *any two* reasons for the great diversity of natural vegetation in India. [3]
- (b) How do the forests help in maintenance of air quality?

### MARKING SCHEME

| (i)                                                                                                                                                                                                                                                                                          | <p>(a) Deforestation, urbanisation, industrialisation, overgrazing, land for agriculture/modernisation/demand for timber or furniture/forest fire./ Construction of multipurpose project/ dam/ construction/ Population pressure/ Mining/ Quarrying/ Shifting agriculture.</p> <p style="text-align: right;">(any two)</p> <p>(b) Afforestation, Agro forestry, social forestry, Van Mahotsav/reafforestation/controlled grazing/creating fire line/silviculture/farm forestry/ Chipko/ Planting trees in different areas.</p> <p style="text-align: right;">(any one)</p>                                                                                                                                                                                                                                                                                                   |                    |                 |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (ii)                                                                                                                                                                                                                                                                                         | <p>Tropical evergreen forest:</p> <p>20-30 degree Celsius</p> <p>150 – 350 cm rainfall/over 200 cms</p> <p>70 – 95% humidity</p> <p style="text-align: right;">(Any one point)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |                 |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                |
| (iii)                                                                                                                                                                                                                                                                                        | <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 50%;">Tropical deciduous</th><th style="width: 50%;">Tropical Desert</th></tr> </thead> <tbody> <tr> <td> <ul style="list-style-type: none"> <li>• Pure stand, easy to exploit, hard durable wood./denser/ economic value/shed their leaves/broad leaves/tall trees/names of trees to be accepted.</li> <li>• Humidity- 50 to 70%</li> <li>• Rain-100-200 cm</li> <li>• Temperature-20-30°C</li> </ul> </td><td> <ul style="list-style-type: none"> <li>• Stunted trees, long roots, waxy leaves, spines, fleshy stem, scattered growth/names of trees to be accepted/xerophytes.</li> <li>• Humidity-Less than 50 %</li> <li>• Rain-Less than 50 cm.</li> <li>• Temperature-20-35°C</li> </ul> </td></tr> </tbody> </table> <p style="text-align: right;">(any three)</p> | Tropical deciduous | Tropical Desert | <ul style="list-style-type: none"> <li>• Pure stand, easy to exploit, hard durable wood./denser/ economic value/shed their leaves/broad leaves/tall trees/names of trees to be accepted.</li> <li>• Humidity- 50 to 70%</li> <li>• Rain-100-200 cm</li> <li>• Temperature-20-30°C</li> </ul> | <ul style="list-style-type: none"> <li>• Stunted trees, long roots, waxy leaves, spines, fleshy stem, scattered growth/names of trees to be accepted/xerophytes.</li> <li>• Humidity-Less than 50 %</li> <li>• Rain-Less than 50 cm.</li> <li>• Temperature-20-35°C</li> </ul> |
| Tropical deciduous                                                                                                                                                                                                                                                                           | Tropical Desert                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |                 |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                |
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| (iv)                                                                                                                                                                                                                                                                                         | <p>(a) Variety of soil, diverse climatic condition, physiography/relief/vast size of the country./ Latitude Extent is vast.</p> <p style="text-align: right;">(any two)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |                 |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                |

- |     |                                                                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b) | Forest give out oxygen / absorb carbon dioxide / Purifies the air./ Carbon sink./ Forest help in bringing rain which in turns improves air quality and control air pollution. |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Comments of Examiners

- (i) (a) This question was accurately answered by most of the candidates.  
 (b) Most of the candidates answered this part of the question correctly.
- (ii) Majority of the candidates answered this question correctly. However, some candidates were confused about temperature and rainfall and, as a result, provided incorrect answers. They were unable to comprehend the specific types and features of forests, as well as the connection between vegetation and climate.
- (iii) A large number of candidates were able to differentiate between tropical deciduous and tropical desert correctly. However, some candidates mentioned 'tropical evergreen vegetation' instead of 'tropical desert'.
- (iv) (a) Majority of the candidates wrote 'the importance of trees' or 'the importance of forests' instead of 'reasons for the great diversity of natural vegetation'.  
 (b) Most of the candidates answered this question correctly.

### Suggestions for teachers

- Build clarity of concept on the reasons for decrease of forest cover in India.
- Encourage students to learn different methods of conservation of forests in India.
- Illustrate in class the various climatic conditions that are linked to vegetation types, using images of different forests and vegetation found in diverse climates.
- Provide regular practice to reinforce students' understanding of temperature and rainfall conditions.
- Revise the characteristics of each natural vegetation belt.
- Guide students to read the question properly.
- Encourage students to learn the reasons for the great diversity of natural vegetation in India.
- Explain to students the role of the forests in human lives.

## Question 7

- (i) Mention *any one* advantage and *any one* disadvantage of **well irrigation**. [2]
- (ii) Suggest *any two* measures to overcome *scarcity of water* in the metropolitan cities of India. [2]
- (iii) Give *one* reason to explain each of the following: [3]
- (a) *Tank* irrigation is **NOT** common in Northern India.
- (b) Canal irrigation is important in Punjab.
- (c) All farmers *do not* use tube well irrigation even if the geographical condition is favourable.
- (iv) State *any three* objectives of rainwater harvesting. [3]

## MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>Advantage:</p> <ul style="list-style-type: none"> <li>Cheap, easy to operate, usage in hand of farmer, can be dug at convenient place, independent source./occupy small area./ Easy to dig. <span style="float: right;"><i>(any one)</i></span></li> </ul> <p>Disadvantage:</p> <ul style="list-style-type: none"> <li>Dries up, manual labour, irrigate small fields/pollution./ Water pollution./ time consuming <span style="float: right;"><i>(any one)</i></span></li> </ul> |
| (ii)  | <p>Rainwater harvesting, recharging ground water, installing water metres, spreading awareness through social media, dug well, bore well, permeable pavement, percolation pit./revival of ancient methods/reuse of water./ watershed management./ Waste water treatment./ Recycling of waste water. <span style="float: right;"><i>(any two)</i></span></p>                                                                                                                          |
| (iii) | <p>(a) Soft soil, large number of rivers/absence of natural depressions/other better sources are easily available./ Absence of hard rock.</p> <p>(b) Perennial rivers, soft soil, agriculture is important/fertile agricultural land to recover the cost/level land or flat plain</p> <p>(c) It is expensive / need cheap electricity/lack of power supply/depletes groundwater <span style="float: right;"><i>(any one)</i></span></p>                                              |
| (iv)  | <ul style="list-style-type: none"> <li>Will help in water conservation.</li> <li>Recharging ground water.</li> <li>Reduces water scarcity.</li> <li>Clean water availability.</li> <li>Checks soil erosion.</li> <li>Reduce flooding/ Reduce surface runoff.</li> <li>Meets the increasing demand for water.</li> <li>Checks groundwater pollution.</li> <li>Supplement water supply during dry period/ Summer.</li> </ul> <p style="text-align: right;"><i>(any three)</i></p>      |

## Comments of Examiners

- (i) Most of the candidates were able to answer the advantages of 'well irrigation' correctly. As for its disadvantages, a few of the candidates mentioned that it cannot be constructed in hilly regions.
- (ii) Almost all the candidates answered this part of the question correctly.
- (iii) (a) A large number of the candidates were able to answer this question correctly.  
(b) Majority of the candidates were able to answer this question correctly.  
(c) Several candidates were able to answer this part of the question correctly.
- (iv) Most of the candidates were able to write three objectives of rainwater harvesting correctly.

### Suggestions for teachers

- Discuss elaborately in class, 'merits and demerits of Well irrigation'.
- Show pictures related to sources of irrigation in parts of India, causes of geographical variation, stories of traditional methods of irrigation, data etc. to generate interest in topics.
- Explain the water crisis faced by India and remedies to overcome it.
- Ask students about the type of terrain necessary for the development of various means of irrigation.
- Discuss and explain the uses of canal irrigation.
- Show a comparative study of different means of irrigation with the advantages and disadvantages of each type.
- Describe the concept of rainwater harvesting with its importance.
- Integrate case studies and data to make discussions meaningful in the geography classroom. For instance, discussing climatic data and the way climate affects life and activities, or discussing irrigation data and the way methods vary from one place to another.
- Focus on the discussions involving quantitative data related to understanding, calculating and interpreting indicators and indices.

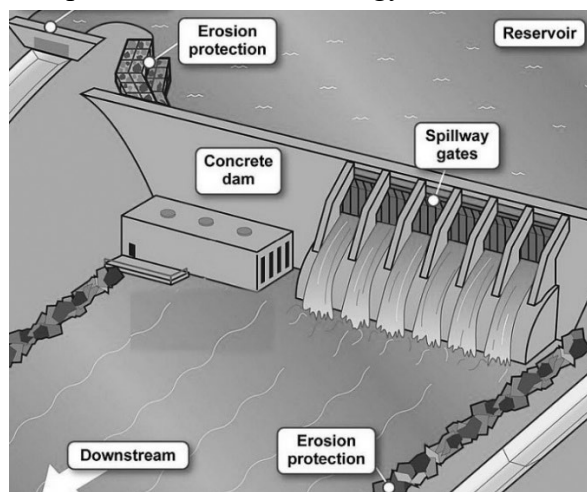
## Question 8

- (i) *With the rising price of fossil fuel every passing day, it is very important to develop non-conventional energy resource.* [2]

— Bulletin – Department of non-conventional sources of energy

- (a) Name a non-conventional energy source that can be harnessed by taking advantage of India's vast coastline.
- (b) State *one* advantage of using *geothermal energy*.
- (ii) (a) How is *Biogas* formed? [2]
- (b) Mention *any one* advantage of Biogas over other non-conventional energy resources.
- (iii) Give a reason for each of the following. [3]
- (a) *Iron ore* is called a versatile mineral.

- (b) *Windmills* set up on agricultural lands are beneficial to farmers.
- (c) *Coal* is the most preferred source of energy in Indian industries.
- (iv) [3]



- (a) Identify the type of energy being generated in the picture given above.
- (b) State why this type of energy is cheaper in the long run.
- (c) Name a power project associated with the generation of this energy in the northern part of India.

### MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>(a) Tidal energy/wind energy</p> <p>(b) Eco friendly / renewable / inexhaustible/sustainable</p> <p style="text-align: right;">(any one)</p>                                                                                                                                                                                                                                              |
| (ii)  | <p>(a) It is the gas obtained by decomposition of vegetative and animal waste.</p> <p>(b) Raw material is easily available, cheap, waste is reduced, extra income./waste of farm is recycled/sludge is used as manure./ Keeps the village clean.</p> <p style="text-align: right;">(any one)</p>                                                                                             |
| (iii) | <p>(a) It is used for making steel which in turn is used for varied purpose – from needle to space craft is made with its help./any use of iron</p> <p>(b) Added income / source of cheap power that is renewable./power may be used for tube well irrigation or for running motor pumps.</p> <p>(c) High calorific value, easily available.</p> <p style="text-align: right;">(any one)</p> |
| (iv)  | <p>(a) Hydel power</p> <p>(b) Less maintenance cost and no expenditure needed for raw material/ low recurring cost/abundantly available/cheaply available./ Renewable/inexhaustible.</p> <p>(c) Bhakra Nangal</p>                                                                                                                                                                            |

## Comments of Examiners

- (i) (a) Majority of the candidates did not pay attention to the question and mentioned solar energy/wind energy, as they skipped to read the word 'vast coastline' as mentioned in the question.  
 (b) Most of the candidates answered this question correctly.
- (ii) (a) Some candidates were not able to explain the process of biogas formation.  
 (b) Almost all the candidates were able to answer this subpart correctly.
- (iii)(a) Majority of the candidates answered this question correctly.  
 (b) Most of the candidates answered this subpart of the question accurately.  
 (c) In this question, several candidates mentioned it accurately; however a few of the candidates mentioned 'the use of coal' instead of 'the energy source that was most preferred' as asked in the question.
- (iv)(a) A large number of candidates were able to identify the type of energy that was generated in the picture.  
 (b) Majority of the candidates answered this part correctly.  
 (c) Most of the candidates mentioned the Bhakra Nangal Dam, while a few did not pay attention to the part of the question that asked "which northern part of India" and thus lost marks.

### Suggestions for teachers

- Explain how relief structures are responsible for deciding the type of non-conventional energy.
- Discuss in class the advantages of non-conventional energy.
- Explain the concept and procedure of biogas formation.
- Guide students to build conceptual clarity so that they can differentiate between other non-conventional energy resources.
- Encourage students to mention the uses of minerals.
- Discuss the advantages of conventional energy resources.
- Give explanations regarding the benefits of multipurpose dams.
- Explain the advantages of hydel energy.
- Build awareness of students towards the names of states benefited by the projects.
- Emphasise issue-based discussion along with the utilisation of various teaching and learning materials, including images, narratives, statistics, and case studies.

## Question 9

- (i) Mention *two* climatic conditions that favour the cultivation of *groundnut*. [2]
- (ii) Give *any two* reasons why *Maharashtra* has emerged as an important *sugarcane* producing state. [2]
- (iii) Give *one* reason to explain each of the following: [3]
- (a) *Trees* are planted in between the rows of *coffee* plants.
- (b) *Cotton* cultivation requires plenty of *cheap labour*.
- (c) *Hill slopes* are best suited for *tea cultivation*.
- (iv) Explain the following: [3]
- (a) Mixed farming
- (b) Zayad
- (c) Transplantation

### MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>Temperature: 20° – 30°C                      Cannot withstand frost                      Around 25/ About 25</p> <p>Rainfall:                      50 - 100 cm                      Around 70/ About 70</p> <p style="text-align: right;"><i>(any two)</i></p>                                                                                       |
| (ii)  | <p>Black soil</p> <p>Tropical climate / maritime climate</p> <p>Temperature: 20 – 30°C</p> <p>Rain: 100 – 200 cm</p> <p>Frost free days</p> <p>Scientifically managed / cooperative farming / capital investment from sugar lobby / Sugar industry located close to plantation./ Absence of loo.</p> <p style="text-align: right;"><i>(any two)</i></p> |
| (iii) | <p>(a) To provide shade / extra income</p> <p>(b) For picking, ginning, manuring. Harvesting</p> <p>(c) Stagnant water is harmful for crop./require well drained soil</p> <p style="text-align: right;"><i>(any one)</i></p>                                                                                                                            |
| (iv)  | <p>(a) Growing crops along with rearing of livestock.</p> <p>(b) Intermediatory/ cropping season between Rabi and Kharif.</p> <p>(c) Growing of rice in nursery and then transplanting the sapling in the main field.</p> <p style="padding-left: 20px;">Growing of crop in nursery and then transplanting the sapling in the main field</p>            |

## Comments of Examiners

- (i) Several candidates mentioned the range of temperature and rainfall incorrectly. In this question, some candidates mentioned only 'one value' instead of the 'range' while other candidates wrote 'vague points like hot climate, dry climate and so on', instead of 'any values'.
- (ii) Most of the candidates managed to write it correctly.
- (iii) (a) Majority of the candidates answered it correctly.  
(b) A large number of candidates were not able to answer this; it seemed like they could not comprehend the question. The candidates either left it or wrote in general about cotton cultivation. A few candidates managed to mention it correctly.  
(c) Several candidates wrote about cool climate on hilltops being a factor that helped to cultivate tea, or that tea leaves were given a unique flavour by the damp cool weather.
- (iv) (a) A large number of candidates wrote mixed farming being a mixture of different crops, or cash and food crops.  
(b) Most candidates wrote correctly; however a few candidates confused it with Kharif crop.  
(c) Majority of the candidates managed to write correctly.

## Suggestions for teachers

- Encourage students to prepare a table for crops, their areas, climatic and soil requirements etc, highlight them if needed, for learning.
- Use a map while teaching, so that students get a clearer idea about the benefits of Maharashtra being beside the sea for growing sugarcane. Correlate it with topics taught earlier, like soil and climate.
- Encourage students to understand and learn instead of rote learning.
- Mention the reason crops like jute, cotton need lot of labour as work cannot be done by machinery.
- Emphasise that the main reason for growing tea bushes on hill slopes is the fact that tea bushes cannot grow in flat land due to stagnant water.
- Explain the three cropping seasons of India, the crops grown and months covered in tabular form.
- Teach the topic of transportation with examples and show actual pictures of transplantation.
- Emphasise the importance of mixed farming being an advantage to farmers, its special importance in India where all farmers keep cattle. Explain its differences with double or multiple cropping and also from silviculture.

## Question 10

- (i) State *any two* factors that are responsible for the growth of the cotton textile industry in Kolkata. [2]
- (ii) '*Karnataka is the leading state for silk production in India.*' [2]  
Give *any two* reasons to justify the above statement.
- (iii) (a) Name the Iron and Steel industry located at Chhattisgarh. [3]  
(b) From where does the industry named by you above get its supply of the following?  
1. Iron ore  
2. Coal

- (iv) Raghav has been given a project by his teacher in which he has to write about the by-products of an industry that is used for preparing shoe polish, carbon paper and cardboard. [3]
- (a) Which industry will he be writing about?
- (b) Name *any two* by-products of the industry mentioned by you.

### MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | Humid climate, port facility, transport network, wide market, cheap labour, power supply, coal from Raniganj or Jharia, soft water from Hooghly./ Capital available./ Skilled Labour.<br><i>(any two)</i>                                                                                                                                                                                                                                                                                                                                       |
| (ii)  | <ul style="list-style-type: none"> <li>• Suitable climate for sericulture / mulberry trees</li> <li>• Wide market</li> <li>• Silk worms reared all through the year</li> <li>• Fresh water available/soft water or alkaline free water</li> <li>• Modern scientific technology</li> <li>• Skilled labour available</li> <li>• Bombyxmori is reared throughout the year</li> <li>• Licenced nurseries</li> <li>• Traditional skill</li> <li>• Sericulture provide employment to women.</li> <li>• Government Subsidy</li> </ul> <i>(any two)</i> |
| (iii) | <p>(a) Bhilai steel plant</p> <p>(b) 1. Iron ore – Dalli Rajhara, Bastar/ Durg</p> <p>2. Coal – Korba, Raniganj, Jharia, Kargati, Bokaro</p>                                                                                                                                                                                                                                                                                                                                                                                                    |
| (iv)  | <p>(a) Sugar industry</p> <p>(b) Bagasse</p> <p>Molasses</p> <p>Press mud</p> <i>(Any two)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Comments of Examiners

- (i) Several candidates mentioned advantages like black soil, raw material availability, etc., as they seemed to relate cotton textile industry with the cultivation of cotton in India.
- (ii) Most candidates managed to get this correct. Only mistake identified was in the spelling of 'Bombyxmori'.
- (iii) (a) Most of the candidates identified the industry as a different one.  
(b) Since the industry identified was wrong, the suppliers were also wrongly identified. At times, however, some candidates managed to identify the correct suppliers, although they made an error in the industry.
- (iv) (a) Most candidates managed to get this part of the question correctly.  
(b) A large number of candidates wrote the products of sugarcane, instead of the 'by-products'.

## Suggestions for teachers

- Mention the difference between the two i.e. agriculture and agro-based industries by using a political map while explaining the chapter.
- Teach agriculture and agro-based industries, simultaneously.
- Encourage students to understand and learn, instead of memorising without thinking.
- Use a map while teaching, pinpoint the locations of the iron and steel factories.
- Insist that students frequently refer to the maps while learning the chapter to remember the locations.
- Mention clearly the difference between products and by-products of certain products in agro-based industries.

## Question 11

- (i) Why is a well-developed transport network important for India? [2]
- (ii) Mention *any two* advantages of *road transport*. [2]
- (iii) Name the following: [3]
  - (a) The mode of transport that ensures comfortable journey, especially for elders and children.
  - (b) The highway connecting *Silchar in Assam* to *Porbandar in Gujarat*.
  - (c) The *cheapest* means of transport for *bulky goods*.
- (iv) Give a reason for each of the following. [3]
  - (a) *Airways* are the most efficient mode of transport to connect *Northeast India* with the rest of the country.
  - (b) *Railways* are easily developed in the *Northern Plains of India*.
  - (c) *Rivers of peninsular India* are **NOT** ideal for *inland water transport*.

### MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | For industrialisation, urbanisation, tourism, agricultural growth, movement of labour.<br>Integrate the nation/promote employment/education/trade/market/ help in economic development./ Rescue and evacuation during emergency.                                                                                                                                                                                                                                                |
| (ii)  | Cheaper, door to door service, flexible, can be made in hilly areas, act as feeder for other means./ Carries Perishable items./ Connect Remote areas Independent/ Available 24 Hours.<br><br><div style="text-align: right;"><i>(any two)</i></div>                                                                                                                                                                                                                             |
| (iii) | (a) Railways<br>(b) Airways<br>(c) East West corridor / expressway/ EW corridor<br>(d) Water transport                                                                                                                                                                                                                                                                                                                                                                          |
| (iv)  | (a) Fast, can cross mountains / forest with ease <span style="float: right;"><i>(any one)</i></span><br>(b) Level land, soft soil, agriculturally important, dense population. <span style="float: right;"><i>(any one)</i></span><br>(c) Rocky surface, rapids and waterfalls are present, seasonal rivers, uneven topography, rivers are not navigable, short and swift rivers./less deep/presence of hidden rocks.<br><div style="text-align: right;"><i>(any one)</i></div> |

## Comments of Examiners

- (i) Most of the candidates attempted this part correctly. However, a few candidates mentioned only one point of the answer, which made them lose a mark.
- (ii) This part of the question was correctly answered by a large number of candidates.
- (iii) (a) This question was correctly answered by the majority of the candidates. However, a few candidates had written Roadways which was an incorrect answer.
- (b) Several candidates gave incorrect answers as the National Highway/North South East West Corridor/Highway.
- (c) A large number of candidates had written correct answers. However, a few candidates wrote incorrect answers, such as 'Railways/Roadways'.
- (iv) (a) This question was correctly answered by the candidates.
- (b) Majority of the candidates were found to be confused in this question, as they had written incorrectly for the 'Northern Plains', as rugged/roads cannot be constructed and so on.
- (c) Most of the candidates answered this part of the question correctly.

### Suggestions for teachers

- Give in class repeated explanations on the way to answer questions.
- Discuss a comparative study of the advantages/disadvantages of different means of transportation.
- Encourage students to make out the differences between the characteristics of all means of transport.
- Guide students to read the question carefully before answering.
- Provide a comparative study of the advantages and disadvantages of waterways.
- Help students to grasp the importance of airways in modern times.
- Discuss the various issues related to the railways in detail
- Provide a clear description of inland water transport.
- Focus more on the types of rivers and their significance for our economic life.

## Question 12

- (i) (a) Give a reason for a substantial rise of E-waste in India. [2]
- (b) What is *dumping*?
- (ii) 'You cannot protect environment unless you empower people, you inform them and help them to understand that these resources are their own, that they must protect them.' [2]
- Wangari Maathai (recipient of Nobel Prize 2004 for Sustainable Development)
- As a responsible student how would you contribute to reduce and manage the waste generated in your surroundings. Explain by giving suitable examples.
- (iii) With reference to the **3Rs**, give a term that best describes each of the following: [3]
- (a) An empty soft drink can is used as a pen stand.
- (b) Old scrap papers are used to manufacture handmade paper.

- (c) No food is wasted in the household.
- (iv) Give a reason for the following statements: [3]
- (a) Organic manure should replace chemical fertilisers.
- (b) There should be a ban on the use of polythene and plastic products.
- (c) Manual handling of waste must be avoided.

### MARKING SCHEME

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | <p>(a) Rising use of electronic gadgets./ Modern Lifestyle/ No effective e-waste management system/ Lack of Proper Literacy about e-waste management. <span style="float: right;">(any one)</span></p> <p>(b) Dumping is a waste disposal method. Waste is collected and taken to large areas on the outskirts of cities where waste is dumped or thrown.</p>                                                                                        |
| (ii)  | <p>Waste may be reused such as old clothes, bottles, cans.</p> <p>Cloth bag should be used to reduce use of polythene / less use of plastic.</p>                                                                                                                                                                                                                                                                                                     |
| (iii) | <p>(a) Reuse</p> <p>(b) Recycle</p> <p>(c) Reduce</p>                                                                                                                                                                                                                                                                                                                                                                                                |
| (iv)  | <p>(a) It is eco-friendly / does not harm the soil or environment / good for growing healthy crops./checks soil pollution./Retain soil moisture./ Checks Biomagnification./ Eutrophication./ Supress Plant Diseases. <span style="float: right;">(any one)</span></p> <p>(b) It leads to pollution / takes long time to decompose./harmful/non-biodegradable <span style="float: right;">(any one)</span></p> <p>(c) It leads to health hazards.</p> |

## Comments of Examiners

- (i) (a) Most of the candidates attempted this part of the question correctly. However, a few candidates incorrectly related it to environmental pollution.
- (b) A large number of candidates were confused between the definition of 'dumping' and that of 'landfills' or 'composting'.
- (ii) A large number of candidates attempted this question correctly; however, a few candidates mentioned only 3 R's without any explanation, while the remaining few were unable to give proper examples, reducing waste from their day-to-day life.
- (iii) (a) This question was correctly answered by the majority of the candidates. A few candidates interchanged answers of iii(a) and iii(b).
- (b) Almost all the candidates correctly answered this question.
- (c) Most of the candidates identified the term correctly.
- (iv) (a) Several candidates did not understand the question properly.
- (b) Majority of the candidates made errors in this part of the question, as they found the language and understanding of the concept difficult.
- (c) A large number of the candidates answered this question correctly.

## Suggestions for teachers

- Familiarise students with different types of pollution.
- Describe in detail the important terms in a more elaborate manner.
- Explain the concept of the 3 R's with examples from our daily life.
- Guide students to read the question carefully.
- Discuss in class the importance of organic fertilizers over chemical fertilizers.
- Explain the concept of Biomagnification and Eutrophication.
- Discuss elaborately in class the health hazards related to manual handling of the waste.
- Encourage students through discussions to comprehend the larger context of environmental and sustainable challenges, actions individuals may take to address them.
- Include debates, discussions, demonstrations of recycling techniques, projects, and field trips to observe recycling first-hand.
- Adopt an issue-based and project-based approach to address contemporary challenges such as population-resource relations, underdevelopment, environmental issues, and biodiversity. Ask students to address current issues and problems and suggest solutions.
- Encourage students to explore different geographical attributes using an interdisciplinary approach, such as physical environment, vegetation, forests and livelihoods, physiography, minerals and the location of heavy industries, etc.

## GENERAL COMMENTS

### Topics found difficult by candidates

- Question 1.i (b-2): Name of the settlement and month.
- Question 1 iii(a): Reason for the presence of water in the reservoir.
- Question 2. (i,ii,v,vii).: Marking and labelling of *Nathu-La pass*, Coastal plains in Kerala, Name of the wind, Standard meridian of India.
- Question 3.(i.): Rainfall
- Question 4.i(b).: Difference between the temperate cyclones and tropical cyclones.
- Question 5 iii (b)(c): Reasoning-based question on red soil and laterite soil.
- Question 5 iv(a)(b).: Formation of black soil and how it retains moisture.
- Question 7.ii. Reason to overcome the scarcity of water in metropolitan cities.
- Question 9 iv(a)(b)(c): Explanation of the terms.
- Question 10 iii(a)(b): Name of the Iron and Steel Industry and supply of minerals.
- Question 11 iv(c): Features of the rivers of peninsular India.
- Question 12 i (b): Definition of Dumping.
- Question 12 ii: Examples for reducing waste in the surroundings.

### Concepts in which candidates got confused

- Six-figure grid reference between 3rd easting and 3rd northing values.
- Definition of contour and contour interval.
- Writing of units.
- Meaning of the terms 'mixed farming', 'Dumping'.
- Formation of 'Bio-Gas'.
- Need for water conservation.
- Writing a term of 3Rs for a given description.

## Suggestions for candidates

- Study the chapters in detail and avoid selective study.
- Follow the scope of the syllabus and cover all the topics instead of selectively studying.
- Try to comprehend concepts better rather than memorising them without understanding.
- Practice map work regularly and refer to the Atlas or Physical Map of India for reference.
- Do regular practice of topographical maps on a regular basis.
- Ask questions in class and get help from the teachers for clarification during lessons.
- Use the technique of colour-coding and highlighting important terms while reading the textbook for better recall of facts.
- Solve ICSE Competency Focused Practice Questions, Specimen papers and previous year's board question paper.
- Read all instructions and questions carefully before attempting them to avoid missing any content.
- Prepare mnemonics, charts, tables, sketch-notes, and revise with their help key definitions, facts and geographical terms.
- Work neatly and write in legible handwriting.
- Use dark pencils for filling maps and label neatly.
- Avoid over-attempting questions in Part II of the paper.
- Differences between questions must be written in tabular form.
- Revise the answer script before submission of the paper.
- Answer the question to the point as per the marking scheme.
- Ensure to mention the keywords while answering.
- Use the 15-minute reading time judiciously to select the answers that can fetch full scores.
- Review your answers before submitting the answer script.
- Remember to mention the units in calculation-based questions.